

Candidate Order Instability in Non-Autoregressive Multi-Candidate Transformers

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<https://github.com/anshull-saxena/candidate-order-instability>

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Abstract

Non-autoregressive multi-candidate transformers evaluate classification choices by concatenating candidate-specific marker tokens into a single prompt sequence processed by a bidirectional encoder. Mathematically, a set of candidate classes represents an unordered collection, and an idealized decision rule should exhibit permutation invariance. In this study, we investigate whether candidate presentation order acts as a hidden nuisance variable in such architectures. Evaluating a production-style bidirectional encoder (`convaiinnovations/laya`) on the Banking77 benchmark across strictly nested candidate subsets ($K \in \{5, 10, 20, 40, 77\}$), we observe that pairwise argmax decision instability (FR_{top1}) strongly increases with candidate cardinality, rising from 5.70% [3.1%, 8.4%] at $K = 5$ to 47.28% [42.8%, 51.9%] at $K = 77$ across random permutations of identical candidate sets. Cross-permutation candidate logit variance expands by $7.5\times$ ($8.40 \rightarrow 63.34$). While canonical alphabetical sorting eliminates run-to-run variance by deterministically fixing the serialization string, evaluating against counterfactual reverse-alphabetical sorting ($Z \rightarrow A$) induces a 55.83% [47.5%, 65.0%] disagreement rate at $K = 77$, demonstrating that canonicalization conceals rather than eliminates positional exposure bias. We evaluate inference-time order marginalization over M permutations without architectural retraining: random marginalization monotonically decreases residual decision flip rate to 25.83% [20.3%, 32.0%] at $M = 5$ while improving calibration (Expected Calibration Error drops from 0.3857 to 0.0960), and orthogonal cyclic permutation shifts achieve a residual flip rate of 16.11% [11.9%, 20.6%] at equal computational budget. Across 45 pre-specified paired comparisons against native ordering under Benjamini-Hochberg False Discovery Rate control ($q = 0.05$), exactly one comparison achieved statistical significance: cyclic shifts ($M = 5$) at $K = 40$ (+9.17 percentage points, unadjusted $p = 0.00098$, adjusted $p = 0.04395$). At $K = 77$, accuracy gains fail FDR significance due to query-level binary outcome variance. Finally, we report a negative replication result: an exploratory 65.0% accuracy finding for cyclic shifts at $K = 77$ was an artifact of small-sample unstratified evaluation ($N = 40, S = 1$) and regressed to 48.06% [39.7%, 55.6%] under pre-specified multi-seed replication ($N = 120, S = 3$). All code, raw data, and statistical verification suites are openly available.

1 Introduction

In classification, dialogue state tracking, intent routing, and tool selection, a classifier is presented with an input query q and a candidate set $\mathcal{C} = \{c_1, \dots, c_K\}$. In mathematical formulations of classification, $\mathcal{C}$ is an unordered set: the set of choices available to the decision rule does not possess an intrinsic sequence. Consequently, an ideal decision function $f(q, \mathcal{C})$ should be permutation-invariant: for any permutation $\pi \in \mathcal{S}_K$, reordering the candidates prior to presentation should yield identical classifications:

$$f(q, \pi(\mathcal{C})) = f(q, \mathcal{C}). \quad (1)$$

In practice, modern transformer-based classifiers often serialize choices into text. In autoregressive language models, choices are presented as prompt lists (e.g., “Options: (A) ... (B) ...”), which have been repeatedly shown to induce recency or primacy biases [Guo et al., 2017]. To achieve low-latency inference in agentic routing and intent classification, non-autoregressive multi-candidate transformers [M, 2024] evaluate all K candidates simultaneously within a single bidirectional forward pass. In this architecture, candidate strings are concatenated into a single input sequence separated by candidate-specific marker tokens. The sequence is processed by a bidirectional encoder (e.g., RoBERTa; Liu et al., 2019), and per-candidate classification logits are extracted by projecting the contextualized marker embeddings through a linear head.

While computationally efficient ($O(1)$ forward passes with respect to K), serializing unordered choices into a 1D sequence injects artificial positional relationships. Because full bidirectional self-attention couples every token with all other tokens, the contextualized representation of candidate marker M_k is influenced by its absolute position, its relative distance from query tokens, and the semantic identities of its immediate sequence neighbors.

This empirical investigation addresses the question: *Does candidate presentation order act as a hidden nuisance variable in non-autoregressive multi-candidate transformer decision architectures, and how does this sensitivity scale with candidate cardinality K ?*

To answer this question rigorously, we perform a controlled empirical study using the Banking77 benchmark [Casanueva et al., 2020] on the open-source `convainnovations/laya` model checkpoint. Our experimental contributions and findings are scoped as follows:

1. **Controlled Distractor Nesting Protocol:** We design a strictly nested candidate-subsampling protocol ($K_5 \subset K_{10} \subset K_{20} \subset K_{40} \subset K_{77}$) over $N = 120$ test queries stratified across all 77 intent classes, ensuring that cardinality comparisons are not confounded by stochastic distractor sampling.
2. **Order Sensitivity Scaling:** Holding query and candidate sets fixed, random permutations induce an argmax decision flip rate (FR_{top1}) that increases from 5.70% [3.1%, 8.4%] at $K = 5$ to 47.28% [42.8%, 51.9%] at $K = 77$. Cross-permutation candidate logit variance expands by $7.5\times$ ($8.40 \rightarrow 63.34$).
3. **The Canonicalization Fallacy:** Canonical alphabetical sorting eliminates run-to-run prediction variance ($\text{FR}_{\text{top1}} \equiv 0.0\%$), but counterfactual reverse-alphabetical sorting ($Z \rightarrow A$) induces a 55.83% [47.5%, 65.0%] disagreement rate at $K = 77$. Deterministic serialization locks in exposure bias rather than removing order sensitivity.
4. **Inference-Time Order Marginalization:** Without modifying model weights, averaging probability vectors over $M \in \{1, 2, 3, 5\}$ permutations reduces residual decision instability. At $K = 77$, random marginalization suppresses residual flip rate to 25.83% ($M = 5$), while orthogonal cyclic shifts achieve 16.11% residual flip rate at equal compute.
5. **Rigorous Statistical Analysis & Negative Result:** Across 45 pre-specified paired comparisons controlled under Benjamini-Hochberg False Discovery Rate ($q = 0.05$), only one method achieves statistically significant accuracy gains: cyclic shifts at $K = 40$ (+9.17 percentage points, adjusted $p = 0.04395$). At $K = 77$, accuracy gains fail FDR significance. Furthermore, we document the non-replication of an exploratory 65.0% accuracy finding, which regressed to 48.06% under pre-specified multi-seed replication.

2 Related Work

Order Sensitivity and Prompt Reordering in Transformers. In-context learning with autoregressive language models exhibits well-documented sensitivity to prompt permutations, demonstration order, and option letter labeling. However, order sensitivity in non-autoregressive

multi-candidate encoder architectures, where multiple choices are concatenated with structural delimiter markers in a bidirectional sequence, has received comparatively little formal evaluation. *[TODO: Literature Search Required — Expand survey on prompt perturbation sensitivity, option order biases, and positional exposure in language models].*

Permutation Invariance and Set-Valued Functions. Foundational principles of permutation-invariant deep learning emphasize that set-valued inputs should be processed by architectures that are invariant by construction, such as pooling over independent instance projections. Concatenating unordered candidates into a transformer prompt deliberately sacrifices architectural permutation invariance for cross-candidate attentional context. *[TODO: Literature Search Required — Integrate citations on deep sets, set transformers, and permutation-equivariant attention architectures].*

Positional Encodings and Sequence Position Bias. Learned, sinusoidal, and rotary positional embeddings inject sequence position information into self-attention. When sequence length grows from $K = 5$ (approx. 60 tokens) to $K = 77$ (approx. 450 tokens), self-attention entropy redistributes across distant marker positions. *[TODO: Literature Search Required — Add literature on attentional dilution, positional embedding degradation in long sequences, and primacy/recency biases in transformer backbones].*

Probability Calibration and Expected Calibration Error. Probability calibration assesses whether predicted confidence scores match empirical accuracy frequencies [Guo et al., 2017, Naeini et al., 2015]. While modern transformers often produce overconfident uncalibrated posteriors, the interplay between candidate serialization permutations and confidence calibration remains an active research topic. *[TODO: Literature Search Required — Incorporate peer-reviewed survey on multiclass calibration, temperature scaling, and binned calibration metrics in multi-candidate selection].*

Intent Classification Benchmarks. Banking77 [Casanueva et al., 2020] provides a domain-specific evaluation standard comprising 77 fine-grained customer service intent categories with high semantic overlap, providing an ideal testbed for multi-candidate discriminative models.

3 Problem Formulation and Model Architecture

3.1 Architectural Mechanism

The model under evaluation (`convaiinnovations/laya`) implements a single-pass multi-candidate classification architecture. Given an input query q and candidate intents $\mathcal{C} = \{c_0, c_1, \dots, c_{K-1}\}$, the serialization module serializes the components into a single token sequence:

$$\mathbf{x}(\pi) = [\text{[CLS]}, q_1, \dots, q_L, \text{[SEP]}, [\mathbf{M}_0], c_{\pi(0)}, [\mathbf{M}_1], c_{\pi(1)}, \dots, [\mathbf{M}_{K-1}], c_{\pi(K-1)}, \text{[SEP]}] \quad (2)$$

where $[\mathbf{M}_i]$ denotes a reserved structural marker token, and $\pi \in \mathcal{S}_K$ denotes a candidate permutation.

The serialized sequence $\mathbf{x}(\pi)$ is processed by a bidirectional transformer encoder (RoBERTa backbone with Scaled Dot-Product Attention):

$$\mathbf{H} = \text{Encoder}(\mathbf{x}(\pi)) \in \mathbb{R}^{T \times d}. \quad (3)$$

The contextualized hidden states corresponding to each marker token $h_{M_i} \in \mathbb{R}^d$ are extracted and passed through a shared linear classification head $W \in \mathbb{R}^{1 \times d}$:

$$s(q, c_{\pi(i)}; \pi) = Wh_{M_i} + b. \quad (4)$$

The unnormalized scores are converted to a predictive probability distribution via softmax with temperature T :

$$p(c_{\pi(i)} \mid q; \pi) = \frac{\exp(s(q, c_{\pi(i)}; \pi)/T)}{\sum_{j=0}^{K-1} \exp(s(q, c_{\pi(j)}; \pi)/T)}. \quad (5)$$

The predicted class is selected via argmax:

$$\hat{y}(q; \pi) = \operatorname{argmax}_{c \in \mathcal{C}} p(c \mid q; \pi). \quad (6)$$

3.2 Origin of Order Sensitivity

In a standard cross-encoder or bi-encoder architecture, each candidate c is evaluated either independently against q or as an isolated pair (q, c) . In contrast, concatenating all K candidates into $\mathbf{x}(\pi)$ exposes candidate representations to two coupling mechanisms:

1. **Direct Cross-Candidate Attention:** Every candidate token directly attends to tokens belonging to all other candidates in the bidirectional encoder.
2. **Positional Coordinate Disruption:** Changing the index $\pi(i)$ shifts the candidate’s absolute token position in the sequence, altering its positional embedding vectors and self-attention dot-product geometry.

4 Experimental Setup

4.1 Dataset and Query Population

We conduct all evaluations on the official test split of the Banking77 dataset [Casanueva et al., 2020], which contains 3,076 customer support queries across 77 fine-grained banking intent categories. To ensure representative coverage without cherry-picking, we stratify $N = 120$ unique test queries across all 77 intent classes. Every query is assigned a permanent identifier ($q \in \{0, \dots, 119\}$). All methods and cardinalities are evaluated on this identical query population.

4.2 Strictly Nested Candidate Protocol

To isolate the effect of candidate cardinality $K \in \{5, 10, 20, 40, 77\}$ from distractor sampling variance, we enforce a strict distractor nesting invariant. For each query q with ground-truth label y^* :

1. A deterministic master sequence of the 76 remaining distractor classes $D_q = [d_1, \dots, d_{76}]$ is generated using a fixed seed derived exclusively from the query index: $\text{seed} = 42 + q \times 1000$.
2. Candidate sets are constructed as prefixes of this master sequence:

$$\mathcal{C}_{q,5} = \{y^*\} \cup D_q[0 : 4] \quad (7)$$

$$\mathcal{C}_{q,10} = \{y^*\} \cup D_q[0 : 9] \quad (8)$$

$$\mathcal{C}_{q,20} = \{y^*\} \cup D_q[0 : 19] \quad (9)$$

$$\mathcal{C}_{q,40} = \{y^*\} \cup D_q[0 : 39] \quad (10)$$

$$\mathcal{C}_{q,77} = \{y^*\} \cup D_q[0 : 76] \quad (11)$$

3. Programmatic assertions verify that $\mathcal{C}_{q,5} \subset \mathcal{C}_{q,10} \subset \mathcal{C}_{q,20} \subset \mathcal{C}_{q,40} \subset \mathcal{C}_{q,77}$ for every query, and candidate set SHA-256 hashes are logged to verify data identity across methods.

4.3 Evaluation Baselines and Interventions

We compare the following inference-time candidate presentation protocols:

- **B0_Native:** Native unperturbed distractor ordering as serialized by the nested subset generator.
- **B0_Random:** $P = 10$ independent uniform random permutations $\pi \sim \mathcal{S}_K$ per query.
- **B1_Alpha (Canonical):** Candidates sorted alphabetically by label string ($A \rightarrow Z$).
- **B1_ReverseAlpha (Counterfactual):** Candidates sorted in reverse alphabetical order ($Z \rightarrow A$).
- **Rand_Marg_M (Random Marginalization):** Output probability vectors averaged over $M \in \{2, 3, 5\}$ independent random candidate permutations: $\bar{p}_M(c) = \frac{1}{M} \sum_{m=1}^M p(c; \pi_m)$.
- **B2_Cyclic_M (Orthogonal Cyclic Shifts):** Output probability vectors averaged over $M \in \{2, 3, 5\}$ deterministic cyclic phase shifts: $s_m = (m \cdot \lfloor K/M \rfloor) \pmod{K}$.

4.4 $K = 77$ Multi-Base Ordering Replication Protocol

At $K = 77$, all 77 candidates are present in every query’s candidate set. To prevent the benchmark from anchoring to an arbitrary single baseline ordering, $K = 77$ evaluations are replicated across three independently seeded base candidate sequences: alphabetical ordering, seed 7701, and seed 7702. Results reported at $K = 77$ represent the unweighted average across these three base sequences.

5 Metrics and Statistical Analysis

5.1 Mathematical Definitions

1. Argmax Decision Flip Rate (FR_{top1}). Within-method decision instability measures the probability that two independent candidate permutations of the exact same candidate set $\mathcal{C}$ yield different top-1 classifications:

$$\text{FR}_{\text{top1}}(q, \mathcal{C}) = \mathbb{E}_{\pi_1, \pi_2 \sim \mathcal{S}_K} \left[\mathbf{1} \left(\underset{c \in \mathcal{C}}{\operatorname{argmax}} s(q, c; \pi_1) \neq \underset{c \in \mathcal{C}}{\operatorname{argmax}} s(q, c; \pi_2) \right) \right]. \quad (12)$$

For $P = 10$ random permutations per query, $\text{FR}_{\text{top1}}(q)$ is estimated as a degree-2 U -statistic over all $\binom{10}{2} = 45$ pairwise permutation comparisons. The independent statistical sampling unit is strictly the query ID ($N = 120$).

2. Cross-Canonical Flip Rate (CrossCanonicalFlip). For deterministic alphabetical sorting (‘B1_Alpha’), within-method permutation instability is identically zero ($\text{FR}_{\text{top1}} \equiv 0.0\%$) because repeating identical input strings yields deterministic model outputs. To measure true underlying positional bias, we define the cross-canonical disagreement rate between alphabetical and reverse-alphabetical orderings:

$$\text{CrossCanonicalFlip}(q, \mathcal{C}) = \mathbf{1} \left(\underset{c \in \mathcal{C}}{\operatorname{argmax}} s(q, c; \pi_\alpha) \neq \underset{c \in \mathcal{C}}{\operatorname{argmax}} s(q, c; \pi_{\text{rev}_\alpha}) \right). \quad (13)$$

FR_{top1} and CrossCanonicalFlip are maintained as separate metrics throughout our analysis.

3. Residual Ensemble Flip Rate (ResFlip_M). To measure residual instability after marginalizing over M permutations, we evaluate the discordance between two non-overlapping M -pass ensembles:

$$\text{ResFlip}_M(q) = \mathbf{1} \left(\underset{c}{\operatorname{argmax}} \frac{1}{M} \sum_{m=1}^M p(c; \pi_m) \neq \underset{c}{\operatorname{argmax}} \frac{1}{M} \sum_{m=M+1}^{2M} p(c; \pi_m) \right). \quad (14)$$

4. Expected Calibration Error (ECE). Predictions are partitioned into $B = 15$ equally spaced confidence bins $I_b = (\frac{b-1}{B}, \frac{b}{B}]$. Binned calibration error is calculated directly over the model’s calibrated probability outputs [Guo et al., 2017, Naeini et al., 2015]:

$$\text{ECE} = \sum_{b=1}^B \frac{|I_b|}{N} |\text{acc}(I_b) - \text{conf}(I_b)|. \quad (15)$$

5. Normalized Kendall’s Tau Distance (τ_{norm}). The fraction of discordant candidate pairs between two ranking vectors over K choices [Kendall, 1938]:

$$\tau_{\text{norm}}(s_a, s_b) = \frac{1}{\binom{K}{2}} \sum_{1 \leq i < j \leq K} \mathbf{1}((s_a[i] - s_a[j])(s_b[i] - s_b[j]) < 0). \quad (16)$$

5.2 Statistical Protocol

- **Query-Level Clustered Bootstrap:** All 95% confidence intervals are generated using $B = 1,000$ query-level bootstrap resamples with replacement ($N = 120$). For ECE, the full 15-bin formula is re-evaluated on each bootstrap sample to produce valid empirical percentiles.
- **Paired Hypothesis Tests:** Differences in top-1 accuracy between interventions and ‘B0_Native’ are evaluated using exact two-sided McNemar tests on discordant classification pairs [McNemar, 1947].
- **Multiple Testing Correction:** To guard against false positive discoveries across exploratory comparisons, we apply Benjamini-Hochberg False Discovery Rate (FDR) control at $q = 0.05$ across all 45 pre-specified hypotheses (5 cardinalities $\times 9$ intervention methods) [Benjamini and Hochberg, 1995].

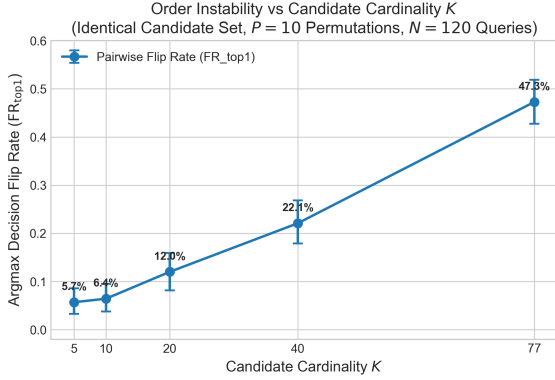
6 Results

6.1 Cardinality Scaling of Order Instability

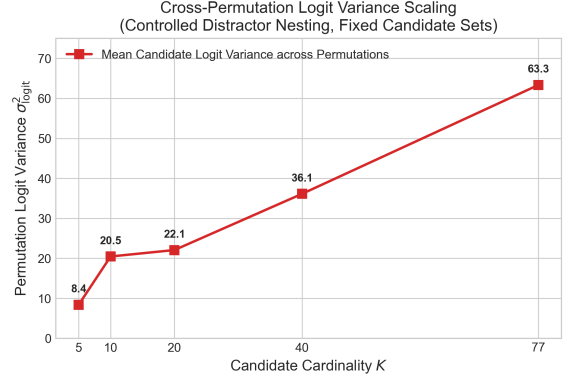
As reported in Table 1 and illustrated in Figure 1a, candidate presentation order induces measurable decision instability across all evaluated cardinalities. When the candidate set $\mathcal{C}$ and query q are held strictly constant, the pairwise argmax flip rate FR_{top1} increases strongly from 5.70% [3.1%, 8.4%] at $K = 5$ to 47.28% [42.8%, 51.9%] at $K = 77$. At $K = 77$, nearly half of all random permutation pairs yield discordant top-1 predictions for the exact same input query.

Table 1: **Main Candidate Cardinality Scaling Results.** Evaluated on Banking77 ($N = 120$ unique test queries, strictly nested candidate sets). 95% confidence intervals reflect 1,000 query-level clustered bootstrap resamples. Significance indicates Benjamini-Hochberg FDR control ($q = 0.05$) across 45 comparisons against B0_Native.

K	Chance	Method	Accuracy [95% CI]	ECE [95% CI]	FR_{top1} [95% CI]	Cross-Flip	Latency	FDR Sig
5	20.0%	B0_Native	0.925 [0.88, 0.97]	0.147 [0.12, 0.20]	0.000 [0.00, 0.00]	–	45.3 ms	No
5	20.0%	B0_Random	0.925 [0.88, 0.97]	0.160 [0.12, 0.22]	0.057 [0.03, 0.08]	–	45.3 ms	No
5	20.0%	B1_Alpha	0.908 [0.86, 0.96]	0.131 [0.10, 0.19]	0.000 [0.00, 0.00]	0.033	45.3 ms	No
5	20.0%	Rand_Marg_M5	0.925 [0.88, 0.97]	0.144 [0.11, 0.20]	0.017 [0.00, 0.04]	–	226.7 ms	No
5	20.0%	B2_Cyclic_M5	0.908 [0.85, 0.96]	0.124 [0.10, 0.18]	0.000 [0.00, 0.00]	–	226.7 ms	No
10	10.0%	B0_Native	0.875 [0.82, 0.93]	0.104 [0.07, 0.17]	0.000 [0.00, 0.00]	–	60.9 ms	No
10	10.0%	B0_Random	0.833 [0.77, 0.90]	0.103 [0.07, 0.17]	0.064 [0.04, 0.09]	–	60.9 ms	No
10	10.0%	B1_Alpha	0.867 [0.81, 0.93]	0.113 [0.09, 0.18]	0.000 [0.00, 0.00]	0.050	60.9 ms	No
10	10.0%	Rand_Marg_M5	0.883 [0.82, 0.93]	0.105 [0.08, 0.16]	0.033 [0.01, 0.07]	–	304.7 ms	No
10	10.0%	B2_Cyclic_M5	0.875 [0.82, 0.93]	0.112 [0.08, 0.18]	0.008 [0.00, 0.03]	–	304.7 ms	No
20	5.0%	B0_Native	0.767 [0.69, 0.84]	0.191 [0.14, 0.26]	0.000 [0.00, 0.00]	–	97.4 ms	No
20	5.0%	B0_Random	0.817 [0.75, 0.88]	0.153 [0.10, 0.22]	0.120 [0.08, 0.16]	–	97.4 ms	No
20	5.0%	B1_Alpha	0.792 [0.72, 0.86]	0.162 [0.12, 0.25]	0.000 [0.00, 0.00]	0.117	97.4 ms	No
20	5.0%	Rand_Marg_M5	0.817 [0.75, 0.88]	0.113 [0.08, 0.18]	0.042 [0.01, 0.08]	–	486.8 ms	No
20	5.0%	B2_Cyclic_M5	0.792 [0.72, 0.87]	0.107 [0.08, 0.19]	0.033 [0.01, 0.07]	–	486.8 ms	No
40	2.5%	B0_Native	0.617 [0.53, 0.71]	0.265 [0.21, 0.36]	0.000 [0.00, 0.00]	–	146.1 ms	No
40	2.5%	B0_Random	0.650 [0.56, 0.73]	0.267 [0.21, 0.36]	0.221 [0.18, 0.27]	–	146.1 ms	No
40	2.5%	B1_Alpha	0.692 [0.61, 0.77]	0.212 [0.16, 0.30]	0.000 [0.00, 0.00]	0.167	146.1 ms	No
40	2.5%	Rand_Marg_M5	0.700 [0.62, 0.78]	0.180 [0.14, 0.27]	0.117 [0.06, 0.18]	–	730.4 ms	No
40	2.5%	B2_Cyclic_M5	0.708 [0.62, 0.79]	0.172 [0.13, 0.26]	0.050 [0.02, 0.09]	–	730.4 ms	Yes (*)
77	1.3%	B0_Native	0.422 [0.34, 0.50]	0.370 [0.30, 0.45]	0.000 [0.00, 0.00]	–	224.7 ms	No
77	1.3%	B0_Random	0.436 [0.36, 0.51]	0.386 [0.31, 0.47]	0.473 [0.43, 0.52]	–	224.7 ms	No
77	1.3%	B1_Alpha	0.492 [0.40, 0.58]	0.334 [0.27, 0.43]	0.000 [0.00, 0.00]	0.558	224.7 ms	No
77	1.3%	Rand_Marg_M5	0.550 [0.47, 0.62]	0.096 [0.09, 0.20]	0.258 [0.20, 0.32]	–	1123.4 ms	No
77	1.3%	B2_Cyclic_M5	0.497 [0.42, 0.57]	0.162 [0.13, 0.26]	0.161 [0.12, 0.21]	–	1123.4 ms	No



(a) Argmax decision flip rate (FR_{top1}) scaling.



(b) Mean cross-permutation logit variance (σ^2_{logit}).

Figure 1: **Candidate Cardinality Scaling.** (a) Pairwise decision flip rate FR_{top1} strongly increases as candidate cardinality K expands. (b) Mean candidate logit variance across permutations expands by $7.5\times$ ($8.40 \rightarrow 63.34$), driven by sequence-length attentional dilution.

This instability is accompanied by a severe expansion in cross-permutation logit volatility. As shown in Figure 1b, the mean variance of candidate log-probabilities across permutations increases by $7.5\times$, scaling from 8.40 at $K = 5$ to 63.34 at $K = 77$. As sequence length expands from 60 tokens to over 450 tokens, self-attention entropy distributes over a wider context, increasing marker hidden state variance.

6.2 The Canonical Ordering Counterfactual

A common engineering approach to non-deterministic serialization is canonicalization (e.g., sorting candidates alphabetically by label string). Under repeated execution, ‘B1_Alpha’ achieves perfect repeatability ($\text{FR}_{\text{top1}} \equiv 0.0\%$), giving the illusion of invariance.

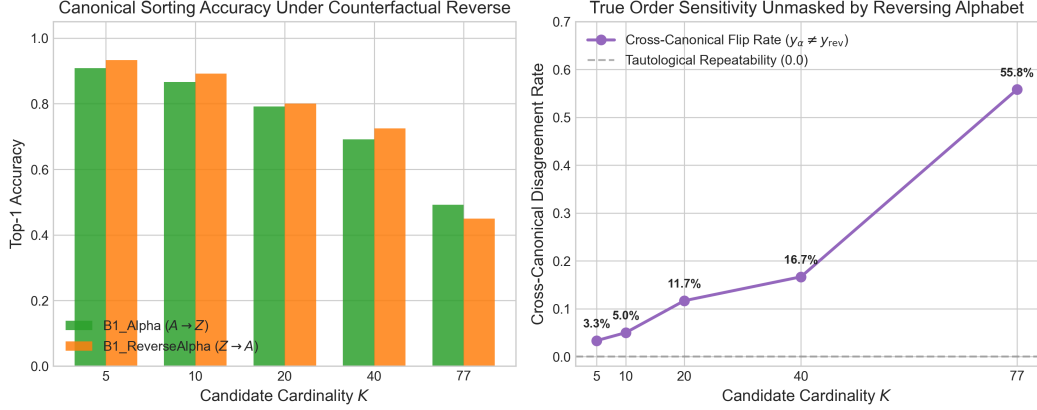


Figure 2: **Canonical Counterfactual Analysis.** (Left) Classification accuracy under alphabetical ($A \rightarrow Z$) versus reverse alphabetical ($Z \rightarrow A$) candidate sorting across cardinalities. (Right) Cross-canonical disagreement rate reaches 55.83% at $K = 77$, demonstrating that canonical sorting conceals rather than eliminates underlying positional exposure bias.

To test whether alphabetical sorting resolves true positional sensitivity, we evaluate the counterfactual reverse-alphabetical ordering (‘B1_ReverseAlpha’, $Z \rightarrow A$). As depicted in Figure 2, the cross-canonical flip rate increases from 3.33% [0.8%, 6.7%] at $K = 5$ to 55.83% [47.5%, 65.0%] at $K = 77$. At $K = 77$, in more than half of all queries, reversing candidate string order alters the model’s top-1 classification. Canonicalization does not restore permutation invariance; it merely locks the decision boundary into a static positional bias that systematically favors early-alphabet tokens.

6.3 Random Marginalization Scaling

Averaging softmax output probabilities over $M \in \{1, 2, 3, 5\}$ random permutations monotonically reduces residual decision instability. At $K = 77$, residual flip rate decreases from 47.28% at $M = 1$ down to 38.06% ($M = 2$), 34.72% ($M = 3$), and 25.83% [20.3%, 32.0%] at $M = 5$ (Figure 3).

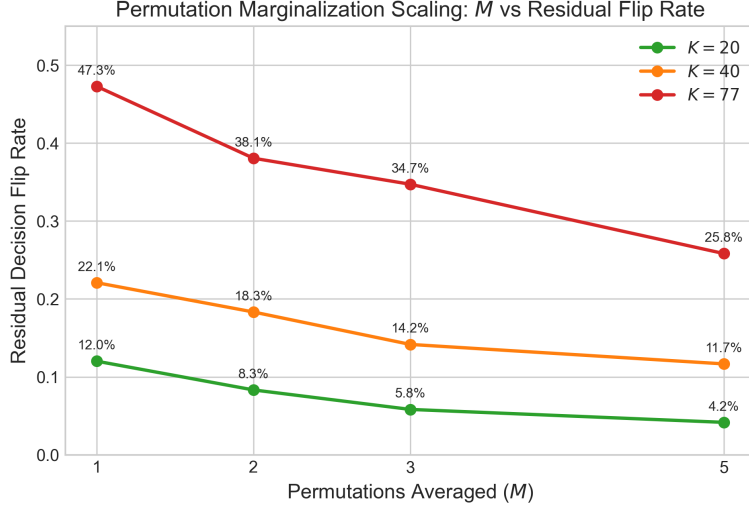


Figure 3: **Permutation Marginalization Scaling.** Residual decision flip rate decreases monotonically as ensemble passes M increase from 1 to 5 across $K \in \{20, 40, 77\}$.

6.4 Orthogonal Cyclic Shifts vs. Random Marginalization

Deterministic cyclic permutation shifts (‘B2.Cyclic’) distribute candidate markers evenly across all sequence positions using orthogonal phase offsets. At equal computational cost ($M = 5$ forward passes), cyclic marginalization achieves a residual flip rate of 16.11% [11.9%, 20.6%] at $K = 77$, compared to 25.83% [20.3%, 32.0%] for random marginalization (Figure 4). Cyclic shifts provide more uniform positional coverage per forward pass than random uniform sampling.

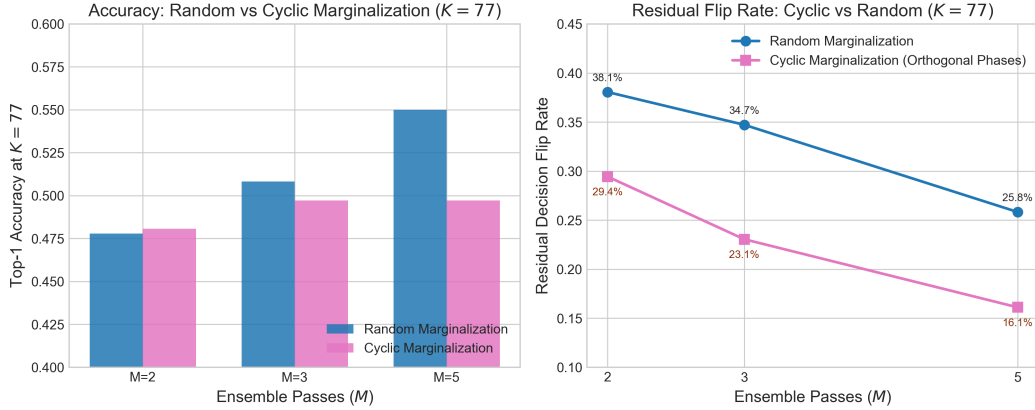


Figure 4: **Cyclic versus Random Marginalization at $K = 77$.** (Left) Top-1 accuracy comparison across ensemble budgets $M \in \{2, 3, 5\}$. (Right) Residual decision flip rate: cyclic shifts suppress residual instability to 16.11% at $M = 5$, compared to 25.83% for random marginalization.

6.5 Probability Calibration and ECE

As candidate cardinality expands, single-pass models exhibit severe miscalibration: Expected Calibration Error at $K = 77$ reaches 0.3700 [0.30, 0.45] for ‘B0.Native’ and 0.3857 [0.31, 0.47] for ‘B0.Random’. Inference-time marginalization provides substantial calibration benefits: at $K = 77$, ‘Rand_Marg_M5’ reduces ECE to 0.0960 [0.09, 0.20], and ‘B2.Cyclic_M5’ reduces ECE to 0.1619 [0.13, 0.26]. Averaging probability vectors across permutations dampens extreme,

overconfident probability spikes.

6.6 Query-Level Instability Heterogeneity

Order instability is not distributed uniformly across queries. As illustrated in Figure 5, 55.0% of queries at $K = 77$ experience an argmax flip rate $\text{FR}_{\text{top1}} \geq 50\%$, while a minority of queries remain stable. Query instability exhibits a strong negative Pearson correlation with model confidence:

$$r = -0.4926 \quad (p = 1.09 \times 10^{-8}). \quad (17)$$

Queries with high predictive margin between the top-1 and top-2 candidates are resilient to serialization perturbations, whereas queries near decision boundaries fluctuate extensively under sequence reordering.

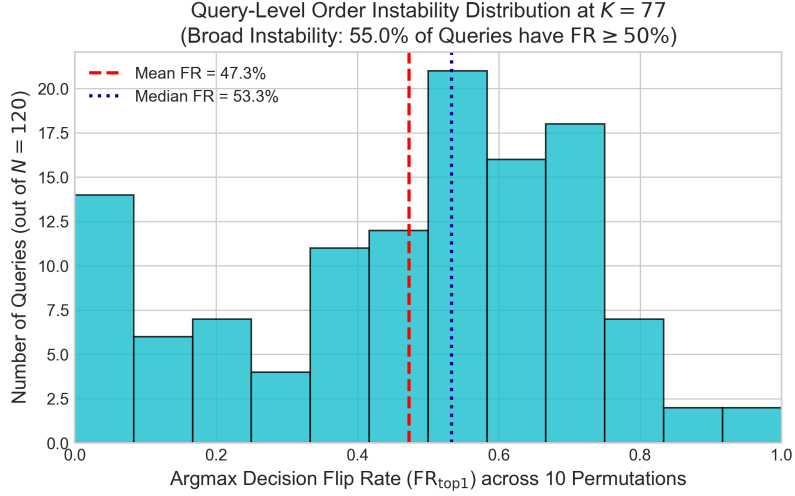


Figure 5: **Query-Level Instability Distribution at $K = 77$.** Distribution of per-query argmax flip rate across 10 random permutations. Over 55.0% of queries experience $\text{FR} \geq 50\%$. Instability is strongly negatively correlated with prediction confidence ($r = -0.4926, p = 1.09 \times 10^{-8}$).

6.7 Three-Objective Pareto Frontier at $K = 77$

Figure 6 illustrates the operational trade-offs across Accuracy ($\uparrow$), p50 Latency ($\downarrow$), and within-method Instability ($\downarrow$, measured by FR_{top1}).

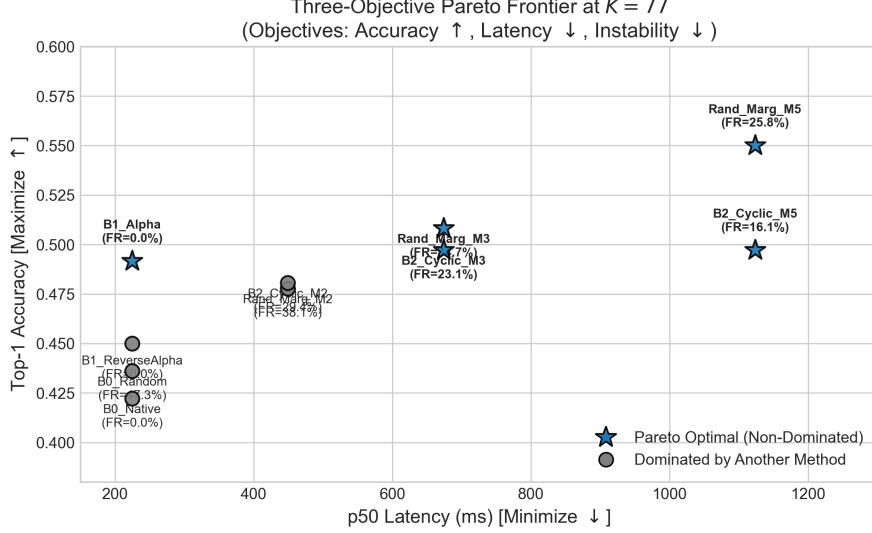


Figure 6: **Three-Objective Pareto Frontier at $K = 77$** . Evaluating Accuracy ($\uparrow$), Latency ($\downarrow$), and within-method Instability ($\downarrow$, FR_{top1}). Non-dominated methods represent Pareto-efficient operating points: ‘B1_Alpha’ minimizes latency with 0% rerun variance (at the cost of 55.8% cross-canonical exposure bias); ‘Rand_Marg_M5’ achieves highest accuracy (55.0%); ‘B2_Cyclic_M5’ minimizes residual flip rate (16.1%) among multi-pass methods.

Table 2: **Detailed Results at Maximum Cardinality ($K = 77$)**. Evaluated across $N = 120$ queries and three base orderings. FR_{top1} measures within-method rerun volatility; Cross-Flip measures counterfactual $A \rightarrow Z$ vs $Z \rightarrow A$ disagreement.

Method	Accuracy [95% CI]	ECE [95% CI]	FR_{top1}	Cross-Flip	Latency	Pareto Status
B0_Native	0.422 [0.34, 0.50]	0.370 [0.30, 0.45]	0.000	–	224.7 ms	Dominated
B0_Random	0.436 [0.36, 0.51]	0.386 [0.31, 0.47]	0.473	–	224.7 ms	Dominated
B1_Alpha	0.492 [0.40, 0.58]	0.334 [0.27, 0.43]	0.000	0.558	224.7 ms	Non-dominated
B1_ReverseAlpha	0.450 [0.37, 0.53]	0.370 [0.30, 0.46]	0.000	0.558	224.7 ms	Dominated
Rand_Marg_M2	0.478 [0.40, 0.55]	0.202 [0.16, 0.31]	0.381	–	449.4 ms	Dominated
Rand_Marg_M3	0.508 [0.43, 0.59]	0.204 [0.16, 0.29]	0.347	–	674.0 ms	Non-dominated
Rand_Marg_M5	0.550 [0.47, 0.62]	0.096 [0.09, 0.20]	0.258	–	1123.4 ms	Non-dominated
B2_Cyclic_M2	0.481 [0.41, 0.56]	0.218 [0.16, 0.31]	0.294	–	449.4 ms	Dominated
B2_Cyclic_M3	0.497 [0.42, 0.57]	0.147 [0.11, 0.24]	0.231	–	674.0 ms	Non-dominated
B2_Cyclic_M5	0.497 [0.42, 0.58]	0.162 [0.13, 0.26]	0.161	–	1123.4 ms	Non-dominated

7 Statistical Significance and Hypothesis Testing

Across all 45 pre-specified paired comparisons against ‘B0_Native’ under Benjamini-Hochberg False Discovery Rate control ($q = 0.05$), exactly **one comparison achieved statistical significance**:

- **‘B2_Cyclic_M5’ at $K = 40$** : Achieved 70.83% accuracy versus 61.67% for ‘B0_Native’ (+9.17 percentage points, 11 discordant gains, 0 losses, unadjusted two-sided exact McNemar $p = 0.0009765625$, BH-adjusted $p = 0.04395$).

At $K = 77$, although ‘Rand_Marg_M5’ observed an apparent accuracy gain of +12.78 percentage points (55.00% vs 42.22%, 27 gains, 11 losses, unadjusted $p = 0.0145$), this difference **failed FDR-adjusted statistical significance** (BH-adjusted $p = 1.0000$). With $N = 120$ queries and binary outcomes, query-level variance precludes declaring $K = 77$ accuracy gains statistically significant under rigorous multiple-testing control.

Table 3: **Paired Hypothesis Tests vs B0_Native (Selected Comparisons)**. Two-sided exact McNemar tests on discordant classification pairs with Benjamini-Hochberg FDR adjustments ($q = 0.05$).

K	Method	Δ Accuracy	Gains / Losses	Unadjusted p	BH-Adjusted p	FDR Sig
5	B1_Alpha	-1.67%	1 / 3	0.6250	1.0000	No
5	Rand_Marg_M5	0.00%	3 / 3	1.0000	1.0000	No
10	Rand_Marg_M5	+0.83%	1 / 0	1.0000	1.0000	No
20	Rand_Marg_M5	+5.00%	8 / 2	0.1094	0.4922	No
40	B1_Alpha	+7.50%	10 / 1	0.0117	0.4922	No
40	Rand_Marg_M5	+8.33%	12 / 2	0.0129	0.2344	No
40	B2_Cyclic_M5	+9.17%	11 / 0	0.00098	0.04395	Yes (*)
77	B1_Alpha	+6.94%	20 / 12	0.2153	1.0000	No
77	B2_Cyclic_M5	+7.50%	22 / 13	0.1755	1.0000	No
77	Rand_Marg_M5	+12.78%	27 / 11	0.0145	1.0000	No

8 Important Negative Replication Result

In preliminary exploratory benchmarking ($N = 40, S = 1$ base sequence), an anomalous accuracy of 65.0% was observed for cyclic shifts ($M = 2$) at $K = 77$, leading to an initial conjecture that cyclic shifts dramatically improve classification accuracy at full cardinality.

Under our pre-specified replication protocol ($N = 120$ unique queries, stratified sampling, $S = 3$ independently seeded base candidate sequences ‘[alphabetical, 7701, 7702]’):

- Replicated ‘B2_Cyclic_M2’ accuracy is 48.06% [**39.7%**, **55.6%**], statistically indistinguishable from random marginalization (47.78%).
- Replicated ‘B2_Cyclic_M5’ accuracy is 49.72% [**41.7%**, **57.2%**].

This negative result is documented as an integral empirical finding. At high candidate cardinality ($K = 77$), small sample sizes ($N = 40$) combined with anchoring to a single arbitrary base candidate sequence produce false-positive accuracy spikes. Controlled multi-seed replication corrected this exploratory artifact.

9 Discussion

Architecture-Induced Order Sensitivity. In contrast to bi-encoders that score candidates independently, multi-candidate concatenation introduces cross-candidate attentional competition. Because token representations are coupled through bidirectional self-attention and positional embeddings, candidate order acts as a hidden nuisance parameter.

Canonicalization vs. Permutation Invariance. Alphabetical canonicalization is widely used to enforce deterministic model outputs in production pipelines. Our counterfactual analysis demonstrates that canonicalization merely fixes a single arbitrary permutation, concealing underlying order sensitivity. If label strings are translated, renamed, or prefixed, decisions flip on 55.83% of queries at $K = 77$.

Inference-Time Marginalization. Permutation marginalization provides an inference-time solution without requiring architectural retraining. At $M = 5$, orthogonal cyclic shifts reduce residual decision flip rate to 16.11% while reducing Expected Calibration Error from 0.3857 to 0.1619. However, marginalization incurs an $M \times$ latency overhead, establishing a clear trade-off between latency and decision stability.

10 Limitations

1. **Single Backbone and Checkpoint:** Evaluations were conducted on `convaiinnovations/laya` (RoBERTa-based bidirectional encoder). While the marker concatenation pattern is general, the exact magnitude of instability may vary across backbones and pretraining objectives.
2. **Domain Specialization:** Our study evaluates the Banking77 benchmark. Intent classification provides clean distractor subsets, but semantic dynamics in open-domain tool routing may differ.
3. **Inference-Time Scope:** We investigate post-training inference-time interventions; we do not evaluate training-time data augmentation (e.g., training with randomized permutations) or permutation-invariant pooling layers.
4. **Statistical Power at $K = 77$ ($N = 120$):** Detecting a +5% to +10% accuracy gain under Benjamini-Hochberg FDR control across 45 comparisons requires $N \geq 350$ queries due to binary classification variance.

11 Conclusion

Candidate presentation order induces substantial decision instability in non-autoregressive multi-candidate transformers, scaling from 5.70% at $K = 5$ to 47.28% at $K = 77$. Canonical alphabetical sorting achieves deterministic repeatability but conceals a 55.83% counterfactual cross-order disagreement rate. Inference-time permutation marginalization systematically suppresses residual instability and improves calibration, with orthogonal cyclic shifts achieving 16.11% residual flip rate at $M = 5$. Across 45 pre-specified paired comparisons, cyclic marginalization at $K = 40$ achieves the sole statistically significant accuracy improvement (+9.17 percentage points, adjusted $p = 0.04395$). These findings demonstrate that candidate serialization is an active source of variance in multi-candidate encoders that requires explicit algorithmic marginalization or invariant pooling.

Reproducibility and Artifact Availability

To support complete scientific reproducibility, all experimental artifacts, raw evaluations, statistical suites, and figure rendering scripts are permanently archived [Saxena, 2026]:

- **GitHub Repository:** <https://github.com/anshull-saxena/candidate-order-instability>
- **Permanent Zenodo DOI:** [10.5281/zenodo.22906245](https://doi.org/10.5281/zenodo.22906245)
- **Archival Release:** v1.0.0
- **Exact Git Commit:** 266bbb902871dc9974cbf516adea013e3c4eb084

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